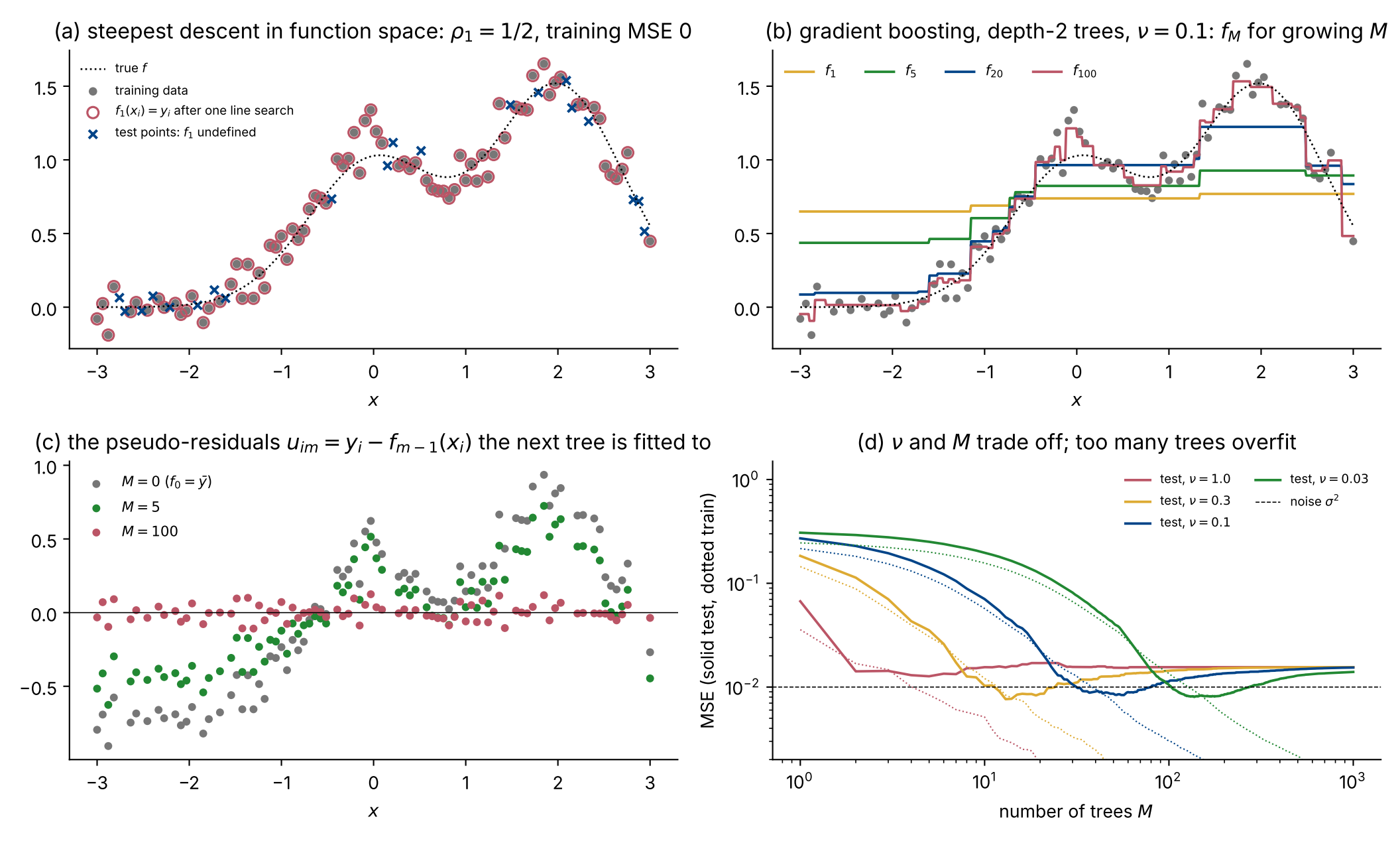
(a) steepest descent in function space: $\rho_1 = 1/2$, training MSE 0
true f
training data
$f_1(x_i) = y_i$ after one line search
test points: f_1 undefined
x
(b) gradient boosting, depth-2 trees, $\nu = 0.1$: f_M for growing M
f_1
f_5
f_{20}
f_{100}
x
(c) the pseudo-residuals $u_{im} = y_i - f_{m-1}(x_i)$ the next tree is fitted to
$M = 0$ ($f_0 = \bar{y}$)
$M = 5$
$M = 100$
x
(d) ν and M trade off; too many trees overfit
test, $\nu = 1.0$
test, $\nu = 0.3$
test, $\nu = 0.1$
test, $\nu = 0.03$
noise σ^2
MSE (solid test, dotted train)
number of trees M